



## K-NFB READER

### What's the K-NFB Reader?

knfbReader is a mobile phone software for the blind, developed by Ray Kurzweil (the father of text-to-speech) in association with National Federation for the Blind.

### How does knfbReader work?

It's a text-to-speech software that takes its input from the mobile phone's camera and then uses optical character recognition technology and text-to-speech capability to read out text. It displays each sentence and highlights each word while it's being read out at a user-adjustable speed. Thousands of pages can be saved in text format using memory sticks. Braille support is available on request.

### Which phone models can sport knfbReader?

Currently, knfbReader is paired with and designed for the Nokia N82, Symbian Series 60, (3rd Edition) platform with 50 MB of flash storage.

### Who's going to benefit from it?

This technology will benefit the visually impaired and those suffering from dyslexia—generally those who need assistance for reading text. It will make them more independent and boost their performance at professional and personal fronts.

### Why will it make a difference?

This mobile reader program is expected to eliminate the need to carry multiple devices and ultimately make the mobile phone a unified solution. The mobile reader can also function with GPS programs and access PDF files. Besides, accessibility to other mobile functions can be improved by using software such as MobileSpeak and Talks, developed specially for the visually impaired.

### When will the knfbReader be available for purchase?

knfbReader was released on February 15, 2008. Get it at <http://www.knfbreader.com/>

gamers to interact with the virtual world using their thoughts and emotions.

Tan Le, President of Emotiv says, "It picks up electrical activity from the brain and sends wireless signals to a computer. It allows the user to manipulate a game or virtual environment naturally and intuitively." Although the concept of reading neural activity is not new, Emotiv claims that the Epoc headset is the first consumer device that can be used for gaming.

Emotiv is also in talks with IBM for developing the technology for use in "strategic enterprise business markets and virtual worlds". According to Paul Ledak, Vice President, IBM Digital Convergence, brain computer interfaces such as the Epoc headset are an important component of the future 3D Internet and the future of virtual communication.

Future 3D Internet! Sounds interesting! India has a perfect blend of the crème de la crème in the high rises of metros and new-to-gaming kind from mid-sized and small cities. While the Internet evolves towards 3D and games increasingly tech-heavy, the pricing and distribution of these gadgets in the Indian market is what we look forward to.

## LIGHT SPEED

# Optical Networks To Blaze Faster

Researchers at the U.S. Department of Energy's Ames Laboratory have come up with an ideal way to sort and distribute data transfer over optical fibres across the globe. The new technology, developed by a team led by physicist Rana

Biswas, is called the Three-dimensional Photonic Crystal Add-drop Filter, and enables enhanced transmission over multiple wavelength channels (wavelengths of light) travelling along optical fibres. This is a significant milestone in the development of all-optical transport networks to eliminate electrical components from optical transmission links, resulting in virtually flawless data reception for end users of the Internet and other fibre-based telecommunication systems.

Biswas explains, "There are up to 160 wavelength channels travelling through an optical fibre at the same time. That means a lot of dialogue is going on simultaneously". Biswas, who is also an Iowa State University adjunct associate professor of physics and astronomy and electrical and computer engineering, explained that as information is transported over these multiple channels, it's necessary to periodically drop-off individual wavelength channels at different points on the fibre. At the same time, it's essential to be able to add data streams into the empty wavelength channels. "When the data being transported in multiple frequency channels over an optical fibre comes to a receiving station, you want to be able to pick off just one of those frequencies and send it to an individual end user," said Biswas. "That's where these 3-D photonic crystals come into play."

To further prove their finding, the researchers used a three-dimensional, microwave-scale photonic crystal constructed from layered alumina rods, containing a full bandgap—a wavelength range in which electromagnetic waves

cannot transmit. Just as electronic bandgaps prevent electrons within a certain energy range from passing through a semiconductor, photonic crystals create photonic bandgaps that confine light of certain wavelengths. The idea of using photonic crystals for add-drop filters is not new. Since the mid 1990s, many groups worldwide have been working to develop the technology with two-dimensional photonic crystals.

"It works," Biswas said, "but there is loss of some intensity to the end user because 2D photonic crystals don't confine the light completely. For example, in a phone conversation, voices would dim out. But with 3D photonic crystal add-drop filters, the communication would be clear."

Although Biswas and his team have shown that 3D photonic crystals would make highly efficient add-drop filters, there are still problems to address. Getting the size of the photonic crystals down to 1.5 microns to work at the wavelengths used for Internet communications is a big challenge. The Ames Lab group now has some of these photonic crystals working in that range, but to make these controlled structures with one input, another output and a defect, required significant amount of work. A future direction is to simplify the design of the add-drop filter by reducing the layers in the photonic crystal—perhaps having all the action happen in one layer.

This comes as a ray of hope to the average Indian surfer, and more so the host of IT and ITES services companies based in India, where ISPs claiming to be broadband providers are not able to provide enough throughput to keep the customer content! ■